

63rd ASH Annual Meeting Abstracts

ORAL ABSTRACTS

722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION

MAIT and V δ 2 Unconventional T Cells Predict Favorable Outcome after Allogeneic HCT and Are Supported By a Diverse Intestinal Microbiome

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Abstract Microbial diversity is associated with improved outcome in patients receiving allogeneic hematopoietic cell transplantation (allo-HCT), however, the mechanism underlying this observation is unclear. Unconventional T cells recognize specific metabolites of bacterial biosynthesis and their role in the post-HCT immunity has not yet been fully clarified. Here we have performed an observational study (n = 174 patients) using 16S rRNA sequencing of early post-HCT patient stool samples (day 7-21 after HCT) paired with multiparameter flow cytometry (performed at day 30 and day 100 after HCT) to explore the relationship between the intestinal microbiome early after HCT and the unconventional T cell populations in circulation. Our

data extend findings of other groups suggesting that mucosal-associated invariant T (MAIT) cells are dependent on a diverse microbiome and are also associated with favorable allo-HCT outcome. In addition, we report for the first time that the V δ 2 subset of $\gamma\delta$ T cells is positively correlated with MAIT cells as well as independent predictors of favorable transplant outcome. We first focused on MAIT cells as these cells respond to metabolites of the bacterial riboflavin biosynthesis pathway and should therefore be responsive to changes in the gut microbiome. MAIT cell frequency on day 30 after HCT was significantly higher in peripheral blood stem cell (PBSC) graft recipients who had higher peri-engraftment stool diversity (day 7-21; $p=0.014$, $n=118$, α -diversity measured using Simpson's reciprocal index, Figure A). Patients with higher-than-median MAIT cell frequency had improved 2-year overall survival ($p=0.047$, $n=118$ PBSC recipients, Figure B) and lower non-relapse mortality ($p=0.031$) compared with patients with lower-than-median frequency. High dimensional flow cytometry analysis using clustering algorithms Uniform Manifold Approximation and Projection (UMAP) and Self Organizing Map (FlowSOM) identified the V δ 2 subset of $\gamma\delta$ T cells as the only differentially abundant population associated with higher MAIT cell frequency, and furthermore, frequencies of these two cell types were highly correlated ($R=0.38$, $p=2.8e-05$, Figure C). V δ 2 cells are activated by the bacterial metabolite, 4-hydroxy-3-methyl-but-2-enyl pyrophosphate (HMBPP), an intermediate of the microbial isoprenoid biosynthetic pathway. V δ 2 cell frequencies were higher in patients with higher stool α -diversity ($p=0.0026$, $n=118$ PBSC recipients, Figure D) and interestingly, appeared protective with regard to acute graft versus host disease (aGVHD), as we observed a higher frequency of V δ 2 cells among patients who had no or grade 1 disease compared with patients who experienced grade 2-4 aGVHD ($p=0.013$).

We next used Linear Discriminant effect Size (LefSE) analysis to identify statistically significant differences between bacterial taxa in our groups of interest (threshold $p>0.01$ and effect size >4) and observed that higher MAIT and V δ 2 cell numbers are associated with higher abundance of bacteria belonging to the phylum *Bacteroidetes* and lower MAIT and V δ 2 cell numbers are associated with higher abundance of the members of the phylum *Firmicutes*. Furthermore, using the PICRUST2 algorithm, which uses 16S amplicon abundance data to predict the abundance of functional pathways, we observed a significantly increased predicted abundance of HMBPP, the phosphoantigen ligand for V δ 2 cells, in the patients with higher circulating V δ 2 cell frequencies ($p=0.00054$).

Our findings confirm our hypothesis that a diverse microbiota supports the reconstitution of protective unconventional T cell populations, namely MAIT and V δ 2 cells, which are in turn associated with a favorable HCT outcome. Although further studies are needed to dissect the functional contribution of these cells to the post-transplantation immune milieu, our work offers a valuable insight into the interplay between the intestinal microbiome and reconstitution of immune subsets after allo-HCT and may aid in the design of microbiota-targeted interventions.

Disclosures **Gomes:** Xbiome: Current Employment. **Zappasodi:** iTeos Therapeutics: Consultancy; Astra Zeneca: Research Funding; Bristol Myers Squibb: Research Funding. **Ponce:** CareDx: Consultancy, Honoraria; Takeda Pharmaceuticals: Research Funding; Generon Pharmaceuticals: Consultancy; Kadmon pharmaceuticals: Consultancy, Honoraria; Ceramedix: Consultancy, Honoraria; Seres Therapeutics: Consultancy, Research Funding. **Giralt:** JENSENN: Membership on an entity's Board of Directors or advisory committees; SANOFI: Membership on an entity's Board of Directors or advisory committees; AMGEN: Membership on an entity's Board of Directors or advisory committees; PFIZER: Membership on an entity's Board of Directors or advisory committees; GSK: Membership on an entity's Board of Directors or advisory committees; JAZZ: Membership on an entity's Board of Directors or advisory committees; BMS: Membership on an entity's Board of Directors or advisory committees; CELGENE: Membership on an entity's Board of Directors or advisory committees; Actinnum: Membership on an entity's Board of Directors or advisory committees. **Peled:** Seres: Other: Intellectual Property Rights, Research Funding and Travel fees; DaVolterra: Consultancy; Other: Other: Jonathan U. Peled had filed intellectual property applications related to the microbiome (referencenumbers #62/843,849, #62/977,908, and #15/756,845); MaaT Pharma: Consultancy. **Perales:** Cidara: Honoraria; Celgene: Honoraria; Merck: Honoraria; MorphoSys: Honoraria; Nektar Therapeutics: Honoraria, Other; Equilium: Honoraria; Karyopharm: Honoraria; Takeda: Honoraria; Bristol-Myers Squibb: Honoraria; Kite/Gilead: Honoraria, Other; Medigene: Honoraria; Novartis: Honoraria, Other; Sellas Life Sciences: Honoraria; Servier: Honoraria; Miltenyi Biotec: Honoraria, Other; Omeros: Honoraria; Incyte: Honoraria, Other; NexImmune: Honoraria. **van den Brink:** Rheos: Honoraria; Merck & Co, Inc: Honoraria; Therakos: Honoraria; Amgen: Honoraria; GlaskoSmithKline: Other: has consulted, received honorarium from or participated in advisory boards; Synthekine (Spouse): Other: has consulted, received honorarium from or participated in advisory boards; Kite Pharmaceuticals: Other; Notch Therapeutics: Honoraria; Nektar Therapeutics: Honoraria; Priothera: Research Funding; Wolters Kluwer: Patents & Royalties; Juno Therapeutics: Other; Frazier Healthcare Partners: Honoraria; DKMS (nonprofit): Other; Seres: Other: Honorarium, Intellectual Property Rights, Research Funding and Stock Options; Forty-Seven, Inc.: Honoraria; Lygenesis: Other: has consulted, received honorarium from or participated in advisory boards; WindMILTherapeutics: Honoraria; Pharmacyclics: Other; Jazz Pharmaceuticals: Honoraria; Novartis (Spouse): Other: has consulted, received honorarium from or participated in advisory boards; Da Volterra: Other: has consulted, received honorarium from or participated in advisory boards; Ceramedix: Other: has consulted, received honorarium from or participated in advisory boards; MagentaTherapeutics: Honoraria; Pluto Therapeutics: Current holder of stock options in a privately-held company, Other: has consulted, received honorarium from or participated in advisory boards.

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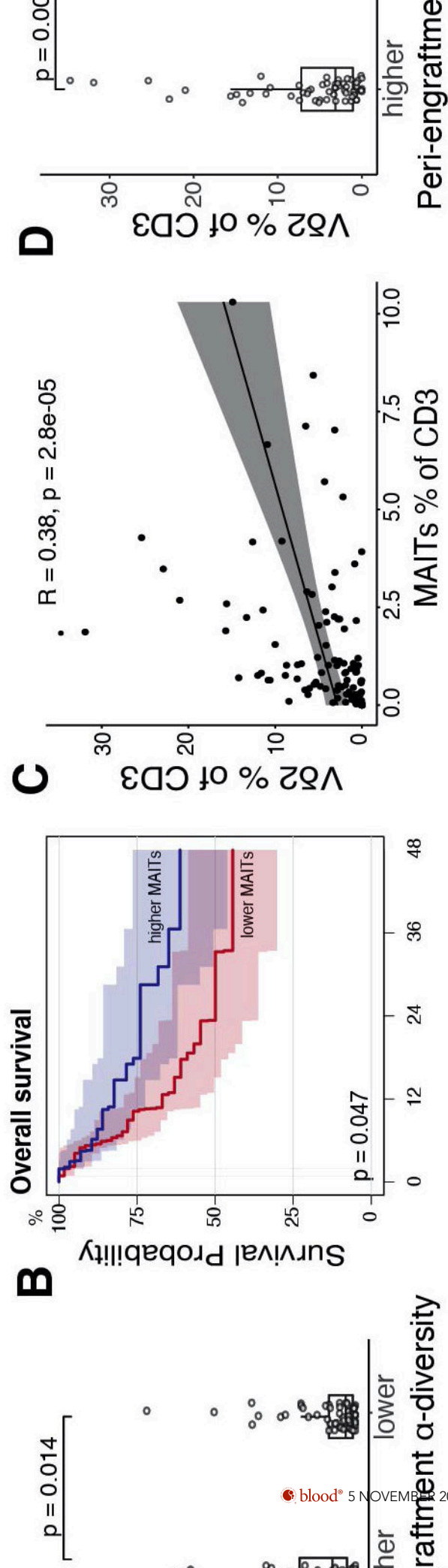


Figure 1